

Fusion of Supervised Feature Selection and Unsupervised Clustering for Multinuclei Classification of MER Signals in DBS for Parkinson's Disease

Xiang Lu^{ID}, Xuexin Du^{ID}, Peng Lun^{ID}, and Yande Ren^{ID}

Abstract—In deep brain stimulation (DBS) surgery for Parkinson's disease (PD), the accurate intraoperative identification of key nuclei—such as the subthalamic nucleus (STN)—is critical to ensuring therapeutic efficacy. However, current approaches heavily depend on expert annotations, and classification tasks are typically confined to binary distinctions between STN and non-STN regions. This limitation hampers the real-time recognition of complex and diverse neuroanatomical structures during PD-related DBS procedures. To overcome this challenge, we propose a three-class classification strategy for microelectrode recording (MER) signals to effectively distinguish the zona incerta (Zi), STN, and substantia nigra (SN). This approach integrates supervised feature selection with unsupervised clustering: discriminative features are first selected using the random forest algorithm; these features are then input into fuzzy c-means (FCM) clustering for preliminary classification; finally, samples with low-confidence scores are manually reviewed. This strategy forms an efficient and verifiable label-generation mechanism that improves classification accuracy and enhances clinical applicability. Experimental results show that the proposed “clustering + review” labeling framework achieves an overall classification accuracy of 92.71%, with only about 10% of samples requiring manual verification—closely matching the 92.97% accuracy achieved with expert labeling and significantly improving labeling efficiency. Furthermore, the ROC-AUC values for all three nuclei (Zi, STN, and SN) exceed 0.97, confirming the model's robust discriminative performance. By combining supervised and unsupervised techniques, the proposed multinuclei classification framework for MER signals not only ensures high accuracy while substantially reducing manual annotation costs but also offers a scalable and efficient solution for rapid neural signal labeling. This method is particularly well-suited for real-time applications such as intraoperative target localization during DBS and shows strong potential for clinical translation.

Index Terms—Deep brain stimulation (DBS), EEGNet1D, fuzzy c-means (FCM) clustering, Parkinson's disease (PD), random forest.

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Xiang Lu and Xuexin Du are with the College of Electronic and Information Engineering, Shandong University of Science and Technology, Qingdao 266590, China (e-mail: sdustlx@sdust.edu.cn; Dxx18863975819@163.com).

Peng Lun is with the Department of Neurosurgery, Affiliated Hospital of Qingdao University, Qingdao 266555, China (e-mail: lunpeng@qdu.edu.cn).

Yande Ren is with the Department of Radiology, Affiliated Hospital of Qingdao University, Qingdao 265000, China (e-mail: 8198458ryd@qdu.edu.cn).

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I. INTRODUCTION

PARKINSON'S disease (PD) is a progressive neurodegenerative disorder primarily characterized by the degeneration of dopaminergic neurons in the substantia nigra (SN) [1]. Its hallmark motor symptoms include resting tremor, muscle rigidity, bradykinesia, and postural instability, all of which significantly impair patients' quality of life [2], [3], [4]. Neuropathological studies have demonstrated that motor dysfunction in PD is closely linked to abnormal neural activity within the basal ganglia circuitry—particularly in the subthalamic nucleus (STN) and the Globus Pallidus internus (GPi) [5]. Currently, medications such as levodopa can be taken to alleviate the symptoms caused by PD [6], [7], [8]. However, long-term drug therapy presents substantial limitations; after 5–10 years of use, many patients experience motor complications such as “wearing-off” episodes and dyskinesias [9]. Moreover, pharmacotherapy fails to halt the underlying neurodegenerative process. For patients with advanced PD who become refractory to medication, surgical intervention emerges as a key therapeutic alternative [10], [11].

Deep brain stimulation (DBS) is a well-established and highly effective neurosurgical intervention that involves implanting electrodes to deliver high-frequency electrical stimulation to specific brain nuclei. This technique modulates abnormal neural circuits, thereby improving motor symptoms and reducing dependence on pharmacologic treatment [12], [13]. Intraoperative microelectrode recording (MER) provides real-time functional guidance for accurate electrode placement [14]. As the electrode advances with submillimeter precision, it captures distinct neural firing patterns from various brain regions (e.g., the STN). By analyzing these electrophysiological signals, surgeons can accurately localize the surgical target [15]. Fig. 1 illustrates the electrophysiological workflow during DBS procedures.

Currently, most MER signal processing methods focus on binary classification between STN and non-STN regions. However, this approach is inadequate for accurately distinguishing adjacent nuclei such as the zona incerta (Zi) and SN within the anatomically complex deep brain structures. Moreover, manual annotation of MER signals is not only time-consuming and highly subjective but also heavily dependent

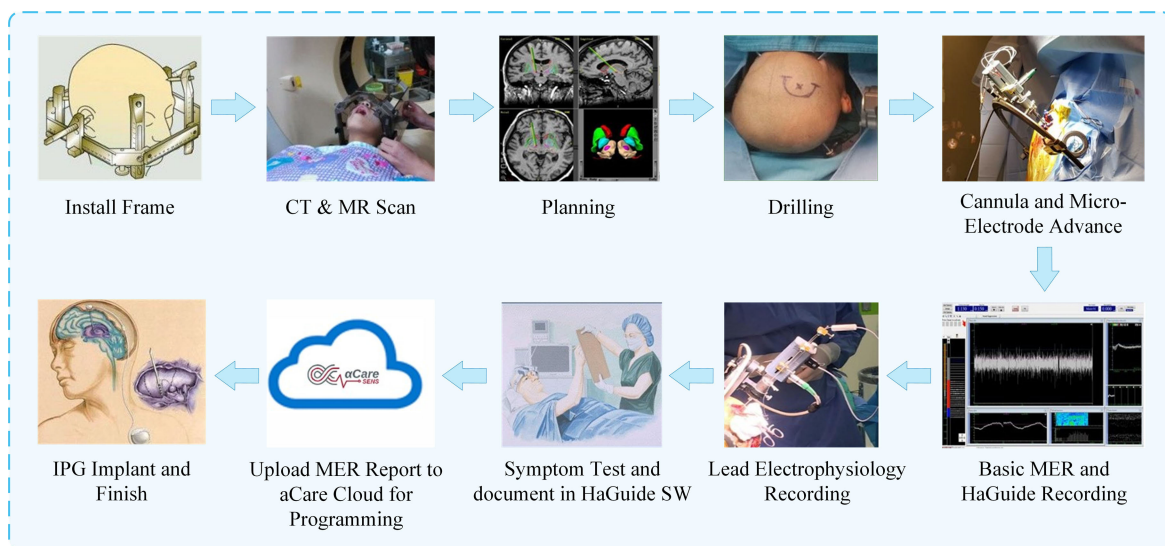


Fig. 1. Electrophysiological workflow of DBS surgery.

on expert experience, limiting its efficiency and scalability in clinical practice [16], [17], [18]. Therefore, there is an urgent need to develop a more refined and automated MER signal recognition method to support high-precision target localization during DBS surgery. Fig. 2 illustrates the electrophysiological signals recorded by microelectrodes as they pass through different nuclei during the procedure. As shown by the anatomical positions of these nuclei, fine-grained and automated classification of MER signals still presents several critical challenges.

- 1) *Ambiguity at Nuclei Boundaries*: MERs acquired along a single trajectory sequentially traverse the Zi, STN, and SN, with substantial overlap in the electrophysiological signatures within boundary (transition) regions. This overlap creates ambiguity in signal attribution. Compared with binary classification, ternary classification requires finer discrimination within these narrow transition zones, placing greater demands on algorithmic robustness and feature sensitivity [19].
- 2) *Redundancy and Noise in High-Dimensional Features*: The 17-D features extracted from MER signals contain redundant and noisy information. Using them directly can degrade clustering purity and impair model generalization. Therefore, feature selection or dimensionality-reduction techniques should be employed to enhance model performance [20], [21], [22].
- 3) *Clinical Reliability of Automated Annotation*: The clustering labels generated by FCM must exhibit strong concordance with anatomical definitions. To improve interpretability and reliability, we combine feature-importance analysis with expert review of low-confidence samples to ensure the biological and clinical validity of the labels.

To address the aforementioned challenges, this study proposes a classification strategy for MER signals that combines supervised feature selection with unsupervised clustering. While retaining the flexibility of unsupervised algorithms, this approach integrates feature selection and label correction

mechanisms to improve classification accuracy and ensure biological consistency. Specifically, a random forest algorithm [23] is first applied to rank the importance of extracted features and select those with high discriminative power. The filtered features are then input into a fuzzy c-means (FCM) clustering algorithm to generate preliminary cluster assignments along with membership scores for each sample. Based on the distribution of these membership scores, a confidence threshold is defined to identify low-confidence samples for manual review, thus establishing a labeling mechanism that primarily relies on automated annotation and is supplemented by expert verification.

Furthermore, to validate the effectiveness of the proposed classification strategy, this study systematically compares classification performance across three types of label sources: fully manual expert-annotated labels, fully automated cluster-generated labels, and hybrid cluster-assisted expert-verified labels. Experimental results demonstrate that although expert labels achieve the highest accuracy, they are labor-intensive and lack scalability. In contrast, cluster-generated labels offer high efficiency but lower precision. The proposed cluster-assisted expert-verified labeling approach significantly increases automation while achieving classification accuracy comparable to that of expert labeling—particularly exhibiting strong robustness in identifying nuclei located near anatomical boundaries. This strategy, for the first time, establishes a closed-loop workflow encompassing feature selection, automated labeling, expert review, and classification validation while also representing the first systematic comparison between unsupervised and expert-derived labeling methods. Fig. 3 illustrates the overall framework for the three-class classification of MER signals and the comparative analysis conducted in this study.

The main contributions of this article are as follows.

- 1) This study proposes a novel strategy for MER signal classification by integrating supervised feature selection with unsupervised clustering. For the first time, it realizes a closed-loop framework encompassing

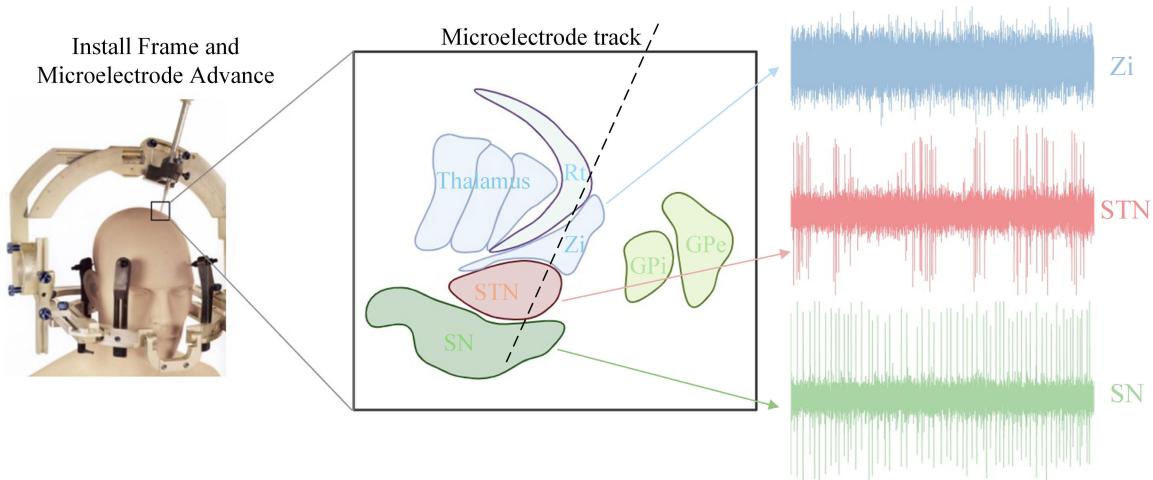


Fig. 2. Electrophysiological signals recorded by microelectrodes as they traverse distinct brain nuclei during DBS surgery.

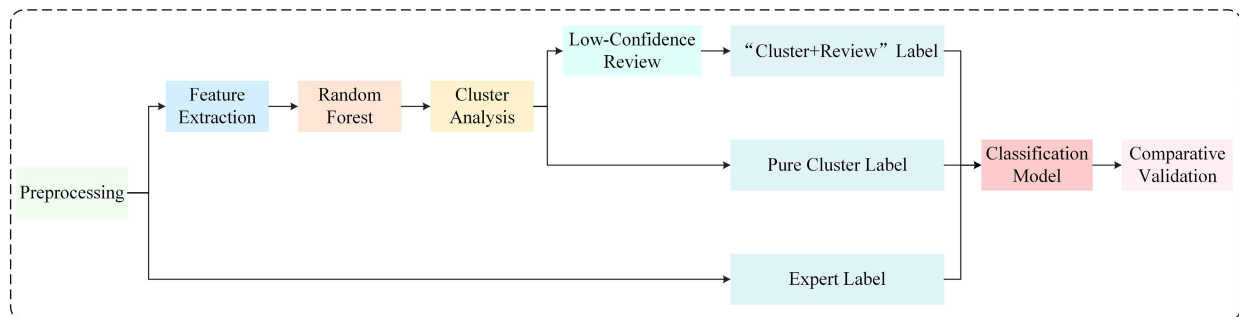


Fig. 3. Overall framework for the comparative analysis of three-class classification of MER signals.

“feature selection—automatic annotation—expert review—comparative validation.” This approach not only ensures high classification accuracy but also significantly enhances the level of automation, effectively addressing the reliance on manual labeling. Moreover, it demonstrates strong potential for clinical application and scalability.

- 2) In this study, the binary classification task is further refined into a three-class classification task, enabling more precise differentiation among the Zi, STN, and SN nuclei. A systematic comparative analysis between expert annotations and clustering-based automatic annotations is conducted for the first time, validating the effectiveness of the approach in terms of recognition accuracy and practical applicability.
- 3) For feature construction and optimization, the random forest algorithm is employed to rank the extracted features by importance, selecting those with higher discriminative capacity. In parallel, a grid search strategy is used to dynamically determine the optimal threshold, providing a robust and reliable feature foundation for the subsequent clustering stage. This process further enhances the overall performance of the classification model.

The remainder of this article is organized as follows. Section II introduces existing methods and models for electrophysiological signal recognition based on supervised and

unsupervised learning. Section III describes the data acquisition process and preprocessing steps, presents the construction of the classification model, and elaborates on the supervised automatic labeling framework, including the design of three control experiments for comparative analysis. Section IV provides a comprehensive analysis of the experimental results. Section V discusses the proposed methodology in detail, along with its practical applicability. Finally, Section VI summarizes the main contributions of this study.

II. RELATED WORK

A. Supervised Learning-Based Identification of Brain Nuclei Using Intraoperative Electrophysiological Signals

Supervised learning methods are widely applied to the classification and brain-nucleus localization of MER signals and related electrophysiological recordings [24]. These approaches rely on manually annotated training datasets, typically mapping segmented MER and local field potential (LFP) signals to corresponding anatomical labels and training discriminative models to support intraoperative auxiliary localization of target nuclei.

From a methodological perspective, existing supervised learning approaches can be broadly categorized into two types: feature-engineering-based classifiers and deep representation learning models. Early—and still widely adopted—schemes emphasize handcrafted features to characterize the statistical

patterns and oscillatory properties of neuronal discharges, combined with traditional classifiers or ensemble learning methods for discrimination. For instance, Benouis and Rosado-Muñoz [25] combined engineered features reflecting discharge activity and LFP-related information with machine learning classifiers to refine STN localization. Similarly, the wavelet packet transform (WPT) is frequently employed to encode MER time–frequency information in a structured manner. Karthick et al. [26] extracted statistical features via WPT and used supervised classifiers to achieve automatic detection of the STN and its adjacent border regions. Furthermore, Hosny et al. [27] extended this approach by combining WPT features with a stacked ensemble strategy. Specifically, they used classical learners as base models and a sequence model as a top-level meta-learner to improve binary classification performance between STN and non-STN. Overall, these studies indicate a trend toward supervised model fusion to address intersubject variability and complex decision boundaries.

Meanwhile, supervised deep learning is driving a shift in brain-nucleus identification from traditional feature-based pipelines toward end-to-end learning. Hosny et al. [28] proposed a CNN–GA–KNN model for STN localization from LFP signals, reflecting attempts to integrate feature learning and downstream classification within a single supervised framework. Beyond pure CNN architectures, attention mechanisms have also been introduced to emphasize more informative segments in time series. Ciecierski and Mandat [29] combined residual networks with temporal attention and aggregated predictions at the window-block level after applying overlapping sliding-window segmentation. Their results indicate that attention-enhanced architectures can improve robustness when signals are partitioned into multiple windows and final decisions require aggregating local evidence. Considering the engineering constraints of intraoperative applications, lightweight networks are also being explored. For instance, Maged et al. [30] proposed a lightweight model that takes spectrograms and scalograms as inputs and integrates residual connections with attention modules. This design aims to reduce computational overhead while maintaining discriminative performance, thereby better aligning with the requirements of real surgical environments.

Although supervised learning methods can achieve strong performance in MER signal classification, their heavy reliance on expert annotations makes model development time-consuming and labor-intensive, limiting scalability to large-scale datasets [31]. Furthermore, in more complex settings such as multinucleus identification, substantially reducing annotation costs without compromising clinical reliability remains a critical challenge.

B. Unsupervised Learning-Based Identification of Brain Nuclei Using Intraoperative Electrophysiological Signals

To mitigate the reliance on expert-labeled data, unsupervised learning approaches have increasingly attracted attention in recent years. These methods do not require prior annotation and can autonomously uncover latent structures or patterns within the input data, thereby substantially reducing the annotation burden for clinical experts [32], [33]. For

instance, Khosravi et al. [34] applied fast Fourier transform (FFT) to extract frequency-domain features from MER signals and utilized a K-means clustering algorithm to classify the signals into STN-IN and STN-OUT categories. This approach enables automatic intraoperative signal grouping without the need for manual labeling and demonstrates good scalability. However, traditional clustering algorithms still face limitations in classification accuracy, as they typically lack effective feature selection and model optimization strategies. In particular, hard clustering techniques such as K-means struggle to handle ambiguous boundaries or overlapping signal characteristics, which can lead to frequent misclassifications. Furthermore, many existing studies lack systematic comparison with expert-labeled annotations, resulting in models with limited interpretability and poor clinical applicability.

III. METHOD

A. Data Acquisition and Preprocessing

The data used in this study were obtained from the Affiliated Hospital of Qingdao University (Qingdao West Coast New Area) and involved patients with PD undergoing DBS surgery. A total of 121 patients were enrolled. For each patient, signals from the left and right hemispheres were collected and stored separately to ensure traceability. In subsequent analyses and dataset partitioning, all data from the same patient were consistently treated as a single subject unit. To obtain evaluation results that better reflect clinical generalization and to avoid potential information leakage across splits caused by overlapping sliding-window segmentation, the dataset was partitioned at the patient level before signal segmentation [35]. Specifically, data from 80 patients were used for training, data from 20 patients were used for validation, and data from the remaining 21 patients were used as the test set for final performance evaluation.

MER signals were recorded using the NeuroNav Physiological Navigation System. This system provides submillimeter positioning accuracy and enables real-time intraoperative recording of raw electrophysiological signals as the microelectrode tip traverses different nuclei. The microelectrode used was the Sterile NeuroProbe (Product Code: STR-007080-00; Manufacturer: Alpha Omega Engineering Ltd.), with a length of 237 mm, an outer diameter of 1.27 mm, and a micro-to-macro distance of 3 mm. The signal analyzed in this study was the CSPK (Continuous Spike) signal, which primarily reflects neuronal discharge activity within the nucleus traversed by the electrode. The sampling rate was set to 24 340 Hz to capture high-frequency neural activity and short-term variations. Each recording session lasted approximately 7–8 s, with the exact duration determined by the electrode dwell time within the target region during surgery.

To enhance signal quality and suppress irrelevant noise, all acquired CSPK signals underwent standardized filtering and segmentation. First, a fifth-order Butterworth bandpass filter was applied to the raw signals with a passband of 300–5000 Hz. The low-frequency cutoff was set to 300 Hz to suppress motion artifacts and power-line interference while preserving the sharp rising edges characteristic of neuronal

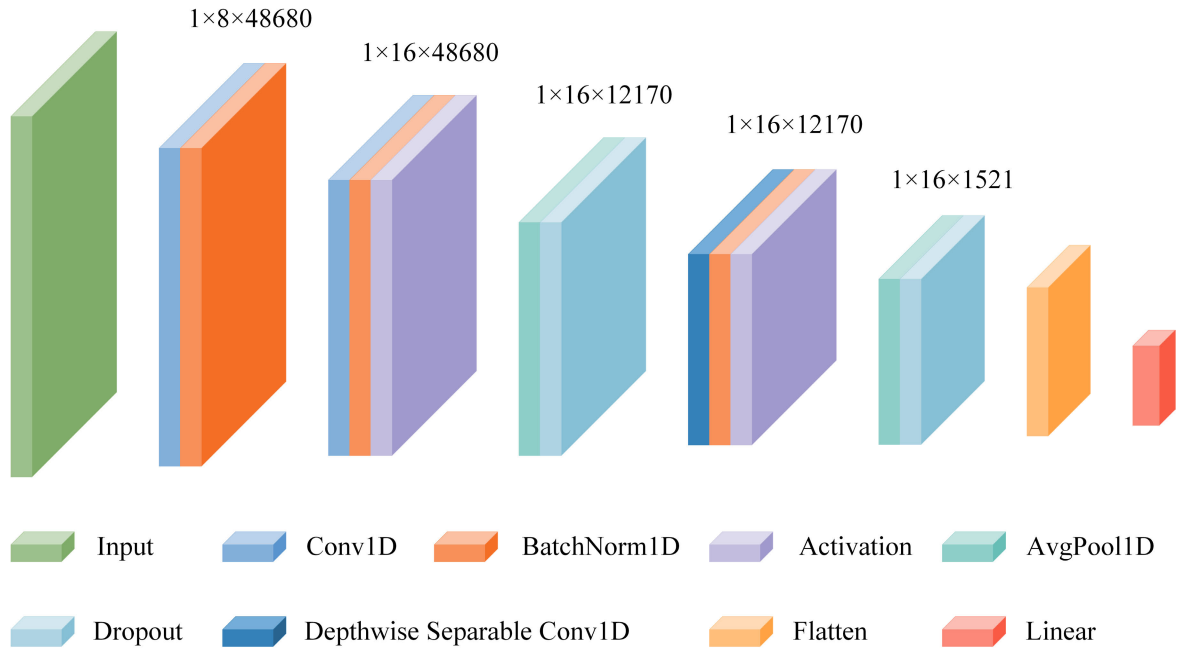


Fig. 4. Network architecture of the EEGNet classification model.

action potentials. The high-frequency cutoff was set to 5000 Hz to attenuate thermal noise and high-frequency interference introduced by the amplifier chain, thereby preventing high-frequency components from masking critical discharge-related features. Subsequently, signals were segmented using a sliding-window scheme, where each segment was 2 s long with 50% overlap (i.e., a 1-s overlap between adjacent segments). This overlap reduces information loss at segment boundaries and preserves temporal continuity, facilitating subsequent feature construction, clustering-based pseudo-label generation, and supervised model training.

B. Architecture of the EEGNet1D-Based Three-Class Classification Model

EEGNet1D [36] is a 1-D lightweight convolutional neural network derived from the EEGNet architecture. It adopts parameter-efficient designs, such as separable convolutions, to reduce computational overhead while preserving temporal representation capability. As a result, it is well-suited for classifying neurophysiological signals in small-sample and noise-contaminated settings. As illustrated in Fig. 4, the network primarily consists of an input layer, a temporal convolution module, a separable convolution module, and a classification head. Specifically, the model first applies 1-D convolutions (Conv1D) to extract local temporal patterns, followed by batch normalization. Channel-wise feature expansion is then performed via pointwise convolutions, together with ELU activations to enhance nonlinear representation. Next, the network proceeds to a depthwise separable convolution module, which comprises depthwise Conv1D and pointwise Conv1D operations applied to each feature map. This design reduces parameter count and computational complexity while maintaining effective temporal feature extraction. After these

core modules, AveragePooling1D is used for downsampling to enlarge the effective receptive field, and dropout is incorporated for regularization to mitigate overfitting. Finally, the extracted high-dimensional temporal features are flattened and fed into a linear classification layer to produce predictions for the Zi, STN, and SN classes.

A key advantage of EEGNet1D is its efficient and lightweight modeling of high-sampling-rate, single-channel MER time series, with strong generalization and practical deployability under small-sample and high-noise conditions. Discriminative information in MER signals primarily arises from spike discharges and their statistics across temporal scales. EEGNet1D leverages 1-D convolutions and hierarchical pooling to form multiscale temporal representations, capturing both local transient patterns and stable differences over longer time intervals. Compared with deeper CNNs or architectures that incorporate complex sequence-modeling modules, EEGNet1D constrains model size through separable convolutions and a compact hierarchy. This helps reduce overfitting when training data are limited, interclass differences are subtle, and noise perturbations are substantial, while improving training and inference efficiency. In summary, EEGNet1D balances temporal feature extraction and generalization stability with controllable computational overhead, facilitating deployment for intraoperative or near-real-time auxiliary analysis. This provides a practical solution for three-class nucleus identification that balances performance and usability.

C. Unsupervised Automatic Labeling Framework

1) *Feature Extraction*: The primary goal of feature extraction is to identify nucleus-specific electrophysiological patterns from intraoperative MER signals. To achieve an accurate interpretation of MER data in patients with PD, this study

extracted a set of time-domain features that reflect both the dynamic changes in neuronal firing activity and the characteristics of the background signal. A total of 17 time-domain features were extracted, comprising 11 spike-related features and 6 spike-independent features. The spike-related features focus on the number, temporal patterns, and burst characteristics of neuronal action potentials, while the spike-independent features capture the overall statistical and morphological properties of the signal.

Spike-related are given as follows.

- 1) *Spike Count (SC)*: The total number of spikes within a signal segment that exceed a predefined threshold

$$SC = \sum_{i=1}^N \mathbb{I}(x_i > T) \quad (1)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function and T is the spike threshold.

- 2) *Spike Firing Rate (SFR)*: The average number of spikes occurring per unit time

$$SFR = \frac{SC}{T_{\text{seg}}} \quad (T_{\text{seg}} = 2) \quad (2)$$

where T_{seg} represents the segment duration (in seconds).

- 3) *Mean Interspike Interval (ISI)*: The average time interval between consecutive spikes

$$ISI = \frac{1}{n-1} \sum_{i=1}^{n-1} (t_{i+1} - t_i) \quad (3)$$

where t_i denotes the time of occurrence of the i th spike.

- 4) *Standard Deviation of ISIs (SISIs)*: A measure of variability in the intervals between spikes

$$SISI = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n-1} [(t_{i+1} - t_i) - ISI]^2} \quad (4)$$

- 5) *Pause Ratio (PR)*: The proportion of time without spike activity relative to the total signal duration

$$PR = 1 - \frac{SC}{f_s \cdot T_{\text{seg}}} \quad (5)$$

where f_s is the sampling rate.

- 6) *Burst Rate (BR)*: The number of bursts per unit time, where a burst is defined as a sequence of consecutive spikes with ISIs less than 60 ms

$$BR = \frac{N_{\text{burst}}}{T_{\text{seg}}} \quad (6)$$

- 7) *Modified Burst Index (MBI)*: The ratio of BR to SFR is used to quantify the burstiness of neuronal activity

$$MBI = \frac{BR}{SFR} \quad (7)$$

- 8) *Burst Total Duration (BTD)*: The sum of durations of all detected bursts

$$BTD = \sum_{i=1}^{N_{\text{burst}}} (t_{\text{end}}^{(i)} - t_{\text{start}}^{(i)}) \quad (8)$$

- 9) *Burst Average Duration (BAD)*: The cumulative duration of all detected bursts within the segment

$$BAD = \frac{BTD}{N_{\text{burst}}} \quad (9)$$

- 10) *Mean Spike Amplitude (MSA)*: The mean amplitude of all detected spikes

$$MSA = \frac{1}{SC} \sum_{i=1}^{SC} |x_{s_i}| \quad (10)$$

- 11) *Standard Deviation Spike Amplitude (SSA)*: The standard deviation of spike amplitudes

$$SSA = \sqrt{\frac{1}{SC} \sum_{i=1}^{SC} (|x_{s_i}| - MSA)^2} \quad (11)$$

Spike-independent

- 1) *Skewness*: A measure of the asymmetry of the signal amplitude distribution is

$$\text{Skewness} = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^3 \quad (12)$$

where N is the number of samples in the segment, x_i is the amplitude of the i th sample, μ is the mean, and σ is the standard deviation.

- 2) *Kurtosis*: A measure of the sharpness of the signal amplitude distribution is

$$\text{Kurtosis} = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^4 \quad (13)$$

- 3) *Average Amplitude Change (AAC)*: The mean absolute difference in amplitude between adjacent sampling points is

$$AAC = \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i| \quad (14)$$

- 4) *Root Mean Square (rms)*: A time-domain representation of the signal's energy intensity is

$$rms = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (15)$$

- 5) *Slope Sign Change (SSC)*: The number of sign changes in the slope of the signal, reflecting waveform variability

$$SSC = \sum_{i=2}^{N-1} \mathbb{I}[(x_i - x_{i-1})(x_i - x_{i+1}) > 0] \quad (16)$$

- 6) *Variance*: A statistical measure of the dispersion in signal amplitude is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \quad (17)$$

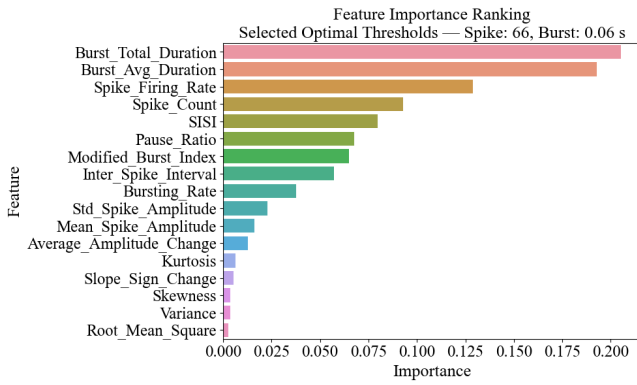


Fig. 5. Feature-importance ranking chart generated by the random forest algorithm.

2) Random Forest for Feature-Importance Analysis:

Random forest, an ensemble learning algorithm, provides feature-importance scores that offer an intuitive measure of variable contributions in medical signal analysis. In this study, we used a random forest to screen and rank multi-dimensional features extracted from MER signals. Using a grid-search-based threshold optimization framework coupled with feature-importance analysis, we enhanced the discriminative capability of features relevant to nucleus classification. To address threshold sensitivity and uncertainty in detecting spikes and bursts in MER signals, we treated the spike and burst thresholds as hyperparameters to be optimized. Accordingly, we constructed a multidimensional feature vector that encompasses spike-related statistical characteristics and burst-discharge indicators. We then performed grid search with threefold cross-validation to optimize the threshold settings used during feature extraction, aiming to identify an effective threshold combination for feature extraction in nucleus classification. After determining the optimal thresholds and extracting features, we fed the resulting feature vectors into a random forest model for training and importance estimation.

Feature importance was quantified using the mean decrease in impurity (MDI). Specifically, MDI measures a feature's contribution by accumulating the weighted decreases in node impurity attributable to that feature across splits in all trees, followed by normalization. Fig. 5 presents the resulting feature ranking. Based on this ranking, we selected several top-ranked features with strong discriminative power for subsequent unsupervised clustering and expert review. Notably, MDI can be biased toward features with many potential split points and may produce unstable importance estimates when features are highly correlated. Therefore, we treat the ranking as a heuristic indicator of feature contribution and use it to derive a more discriminative feature subset for subsequent clustering and review. In summary, this random-forest-based feature selection strategy improves feature-level discriminability and provides a more robust feature foundation for intraoperative localization of the STN in patients with PD.

3) *FCM Clustering and Automatic Labeling Strategy:* This study employed FCM clustering to perform soft clustering on MER signals and to generate pseudo-labels. To ensure subject-level independence in the evaluation protocol, the

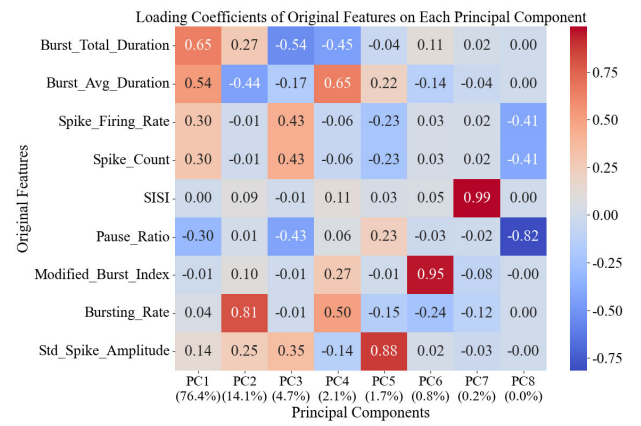


Fig. 6. Heatmap of loading coefficients of original features on each principal component. The X-axis denotes the principal components, with each component annotated by its explained-variance ratio, indicating the proportion of the total variance captured by that component. The Y-axis lists the original features. Each cell in the heatmap indicates the loading coefficient of a specific feature on a particular principal component. The loading coefficient quantifies the strength and direction of the correlation between a feature and the corresponding component: a positive value denotes a positive correlation, while a negative value indicates a negative correlation. The greater the absolute value of the loading coefficient, the stronger the association between the feature and the principal component.

unsupervised clustering pipeline—including feature normalization, PCA-based dimensionality reduction, and clustering hyperparameter selection—was conducted exclusively on the training set (80 patients). Specifically, clustering was completed on the training set and the corresponding membership matrix was estimated. The validation set and the independent test set were excluded from this process and were used solely for subsequent performance evaluation and comparative analysis based on expert annotations. To ensure scale consistency of clustering features, all features were first normalized to the [0, 1] range using MinMaxScaler. Subsequently, based on the feature-importance ranking produced by the random forest, nine highly discriminative features were selected and reduced in dimensionality via principal component analysis (PCA). This step aimed to reduce feature dimensionality, remove redundant information and noise, and facilitate subsequent visualization and interpretation of clustering results. To improve interpretability in the reduced space, we further computed the loading coefficients of each original feature on each principal component and the contribution ratio of each original feature to the principal components. These quantities were visualized as heatmaps to support subsequent feature interpretation and biological-significance analysis (Figs. 6 and 7).

In the reduced principal component space, the FCM algorithm was applied to perform soft clustering. A parameter grid was defined, and an exhaustive search was conducted over key hyperparameters, including the number of clusters c , the fuzziness coefficient m , the error tolerance (error), and the maximum number of iterations (maxiter). The silhouette coefficient was used as the clustering-quality metric to select the optimal parameter combination. After determining the optimal parameter settings, all pairs of principal components were further traversed to automatically identify the pair yielding the

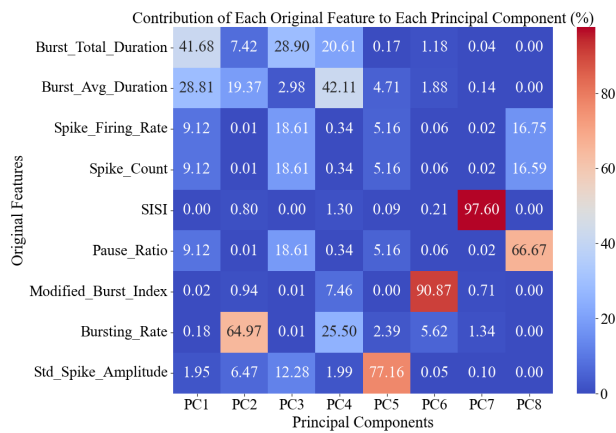


Fig. 7. Heatmap of percentage contribution of original features to each principal component. Each cell represents the relative percentage contribution of a specific feature to a given principal component, calculated as the square of the normalized loading coefficient. Higher values indicate greater contributions of the feature to the corresponding principal component. Features with higher contribution ratios are considered to be major contributors in shaping that component.

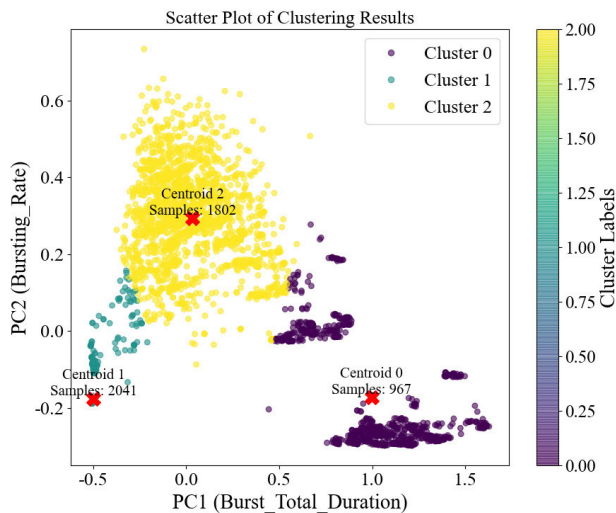


Fig. 8. Clustering scatterplot after PCA dimensionality reduction. The 2-D visualization is based on the two principal components with the highest contribution rates, where Burst_Total_Duration and Bursting_Rate are the original features with the greatest contributions to PC1 and PC2, respectively. The three different colors represent three distinct clusters. Red markers indicate the cluster centroids, with the number of samples labeled next to each centroid. This figure facilitates the validation of the clustering performance, reveals category distributions and ambiguous boundary regions, and provides a reference for subsequent manual review and error analysis.

highest silhouette coefficient, which served as the basis for the optimal 2-D visual projection of the clustering results. Based on this optimal principal component pair, the final clustering map was generated to facilitate intuitive inspection of cluster boundaries and distribution characteristics (Fig. 8).

After clustering, the resulting membership matrix records each sample's degree of membership to each cluster center. Following the maximum-membership criterion, each sample was preliminarily assigned to the cluster with the highest membership value, yielding a deterministic clustering output. To enable automatic labeling and subsequent expert review, a confidence-based label assignment strategy was adopted.

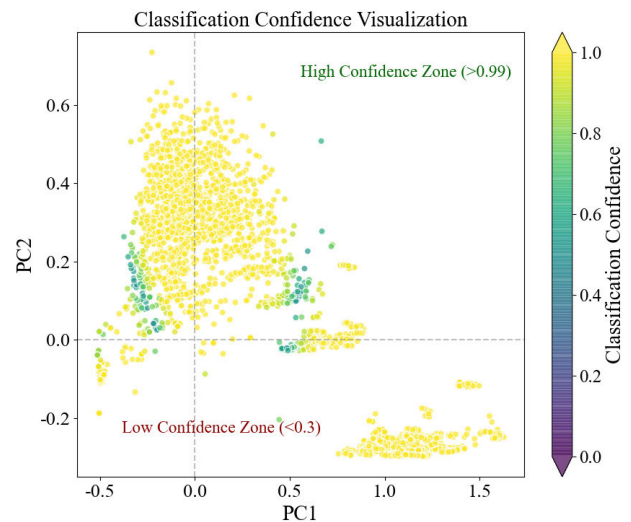


Fig. 9. Confidence distribution plot based on membership degrees. Points with a membership degree greater than 0.99 are considered high-confidence samples and can be directly used for intraoperative decision-making. Points with membership between 0.3 and 0.99 are regarded as medium-confidence samples, which will trigger an expert review mechanism to ensure classification accuracy. As shown in the figure, low-confidence samples are primarily distributed near the boundaries of different clusters, indicating higher uncertainty in these regions. This figure not only provides a quantitative evaluation of clustering confidence but also serves as an important reference for subsequent manual review and classification refinement.

Specifically, a confidence threshold range of 0.30–0.99 was defined. Samples with membership values at or above the upper threshold (≥ 0.99) were treated as high-confidence and assigned to the definitive set. Corresponding folder structures were automatically created according to their assigned cluster labels, and the raw signal files were saved into the respective category folders. Samples with membership values within the intermediate range (0.30–0.99) were treated as requiring manual review; these samples were saved to a separate folder, and a review report was automatically generated. The report details each sample's cluster label and membership information to facilitate subsequent manual intervention and label correction. Samples with membership values at or below the lower threshold (≤ 0.30) were regarded as noise and discarded. Finally, a confidence distribution plot was generated according to the defined thresholds (Fig. 9) to visualize overall membership levels, providing a reference for assessing clustering reliability and guiding manual review. After manual review, the system automatically reads the updated results and replaces the corresponding cluster labels, reassigning the samples to the target clusters. This establishes a closed-loop framework that integrates automated clustering with expert feedback.

D. Validation of Equivalence

To evaluate the effectiveness and practicality of the proposed automatic annotation framework for MER-assisted target identification, we designed three controlled experiments: an expert-labeled group, a pure clustering-labeled group, and a hybrid clustering–revision-labeled group. All experiments adopted a patient-level data partitioning strategy to ensure subject independence across the training, validation, and test

sets. To ensure reproducibility when comparing labeling strategies, training hyperparameters were optimized solely based on expert-consensus labels. Model training was conducted on the training set, and validation-set performance was used to determine the optimal configuration. This configuration was then applied uniformly across all three labeling strategies to enable an attributable comparison. The optimal training iteration was selected during training based on validation-set performance. Unsupervised clustering procedures were performed exclusively on the training set. Final performance metrics were consistently reported on an independent test set, using expert-consensus labels as the ground-truth reference to quantify the consistency between different label-generation strategies and expert annotations.

1) *Expert-Labeling Group*: This group served as the baseline control. Expert labels were jointly annotated by two clinical specialists experienced in intraoperative electrophysiological interpretation. For samples with discrepant annotations, the final label was determined by consensus through discussion, and the resulting expert-consensus label was treated as the ground truth. Using these expert-consensus labels, a supervised classification model was trained on the training set, with hyperparameter tuning performed on the validation set. The trained model was then evaluated on an independent test set, using expert-consensus labels as the reference for subsequent comparisons with the other two label-generation strategies.

2) *Pure Cluster Labeling Group*: This group was designed to evaluate whether pseudo-labels generated under fully automated conditions can be used to train classification models that are highly consistent with expert-defined nuclei. Importantly, the clustering pipeline was conducted solely on the training set. Specifically, feature normalization, PCA-based dimensionality reduction, and clustering hyperparameter selection were all performed exclusively on the training data. FCM clustering was then applied to the training set to obtain membership values for each sample, and pseudo-labels were assigned using the maximum-membership criterion. These clustering-derived pseudo-labels were used to train the same supervised classification model. With training hyperparameters fixed, the optimal training iteration was selected based on validation-set performance. Final evaluation was conducted on an independent test set using expert-consensus labels as the ground-truth reference, in order to quantify the consistency between clustering-generated training labels and expert annotations.

3) *Mixed Labeling Group*: This group evaluated the trade-off between annotation cost and label quality by improving pseudo-label quality through selective expert intervention. Consistent with the pure cluster labeling group, all unsupervised steps were performed solely on the training set. After clustering, samples were stratified by confidence according to their membership values: high-confidence samples retained their cluster labels, whereas low-confidence samples were submitted to experts for review and correction. The resulting hybrid label set comprised automatically assigned cluster labels for high-confidence samples and expert-reviewed labels for low-confidence samples, which were then used to train

supervised classification models. With training hyperparameters fixed, the optimal training iteration was selected based on validation-set performance. Finally, all labeling strategies were evaluated on an independent test set against expert-consensus labels as the ground-truth reference, enabling an objective assessment of label quality and agreement across different annotation sources.

IV. EXPERIMENTS AND RESULTS

A. Experimental Parameter Details, Evaluation Metrics, Classification, and Clustering Baseline

1) *Experimental Parameter Details*: All experiments were implemented in PyTorch and trained on NVIDIA RTX A6000 GPUs with CUDA acceleration. The classification task involved three classes: Zi, STN, and SN. The inputs were single-channel MER signal segments of 2-s duration sampled at 24 340 Hz, corresponding to 48 680 samples per segment.

The number of filters in the first temporal convolution layer was set to $F1 = 8$ with a kernel length of 487, followed by pointwise convolutions for feature expansion. The depth multiplier was set to $D = 2$. Downsampling was performed via two stages of 1-D average pooling with strides of 4 and 8, respectively, to enlarge the effective receptive field and reduce the feature length. Separable convolutions were implemented using depthwise Conv1D and pointwise Conv1D: the number of groups in the depthwise Conv1D was set to the number of input feature maps, and the number of output feature maps in the pointwise Conv1D was set to $F2 = 16$. To mitigate overfitting, dropout was applied after the pooling stage with a dropout rate of 0.5. Finally, the temporal features were flattened and fed into a linear classification layer to produce three-class predictions. Max-norm regularization was applied to the classification layer weights with a constraint threshold of 0.25 to improve training stability and generalization.

For optimization, the Adam optimizer was used with a learning rate of 1×10^{-3} and a batch size of 64 for 80 epochs. The cross-entropy loss was adopted, and class weights were applied according to the class distribution in the training set to mitigate bias induced by class imbalance.

2) *Evaluation Metrics*: This study aims to validate the reliability and discriminative power of labels generated by different annotation strategies. Using a unified EEGNet1D training setup, we evaluate classification results from multiple perspectives with the following metrics: confusion matrix, ROC curve and AUC, accuracy, precision, recall, and $F1$ -score. Taken together, these metrics assess overall performance, interclass discriminative capability, and error types, providing a quantitative basis for objectively comparing different annotation strategies.

3) *Classification and Clustering Baseline*: To validate the effectiveness of the proposed method and to support the rationale for model selection, we established two types of baselines—classification baselines and clustering baselines—under a unified data partitioning and evaluation protocol. All clustering-related steps were performed exclusively on the training set, and final performance was uniformly evaluated on an independent test set using expert-consensus

labels as the ground-truth reference. The validation set was used for model selection and for determining the optimal training epoch, ensuring reproducibility and comparability of results.

Within the expert-labeling group, EEGNet1D was adopted as a 1-D temporal sequence classifier, and two classical machine learning methods were included as strong baselines: SVM and KNN. All three models were compared under identical data partitions and consistent evaluation metrics to quantify the performance gains of EEGNet1D over traditional classifiers and to analyze its discriminative behavior for nuclei that are prone to confusion (STN and SN).

In the pure cluster labeling group, FCM was used as the primary unsupervised labeling method, and spectral clustering was included as a clustering baseline for comparison. Spectral clustering produces hard partitions by assigning each sample to a single cluster, yielding a discrete integer cluster ID as the label. Final evaluation was conducted on an independent test set using expert-consensus labels as the ground-truth reference.

In the hybrid labeling group with cluster review, Gaussian mixture models (GMMs) and HDBSCAN were introduced as clustering baselines for comparison. For all three methods, clustering and parameter selection were performed solely on the training set. Samples were then divided into two subsets according to confidence scores: those accepted directly and those requiring expert review, thereby forming a hybrid training label set. Specifically, GMM outputs posterior probabilities over Gaussian components for each sample; the component with the maximum posterior probability was assigned as the integer label, and this maximum posterior probability was used as the confidence metric for subsequent hierarchical review. HDBSCAN generates discrete cluster identifiers and can identify low-density samples; initial integer labels were constructed based on its cluster assignments, and low-confidence or noisy samples were incorporated into the review process. Finally, a unified comparison was performed on an independent test set using expert-consensus labels as the ground-truth reference, to objectively assess the tradeoff between reducing annotation cost and maintaining label quality across different clustering baselines.

B. Supervised Training With Expert-Consensus Labels

Under the expert-consensus labeling scheme, EEGNet1D was trained in a supervised manner on the training set, with hyperparameters tuned on the validation set, and performance ultimately reported on an independent test set. To quantify the benefits of deep temporal models over traditional classifiers, SVM and KNN were included as supervised baselines for comparison under identical data partitioning and consistent evaluation metrics.

As shown in the confusion matrix in Fig. 10, EEGNet1D produced a clear diagonal pattern on the independent test set, indicating overall strong agreement between predictions and ground-truth labels. The Zi class exhibited the most stable recognition, with 455 samples correctly classified; only seven samples were misclassified as STN, and none as SN. STN classification was generally strong, although some confusion remained, mainly involving 27 samples misclassified as SN

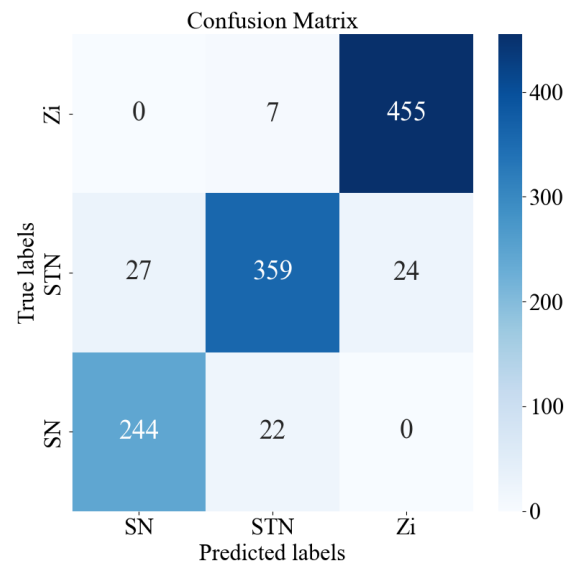


Fig. 10. Confusion matrix of EEGNet1D trained with expert labels on the test set.

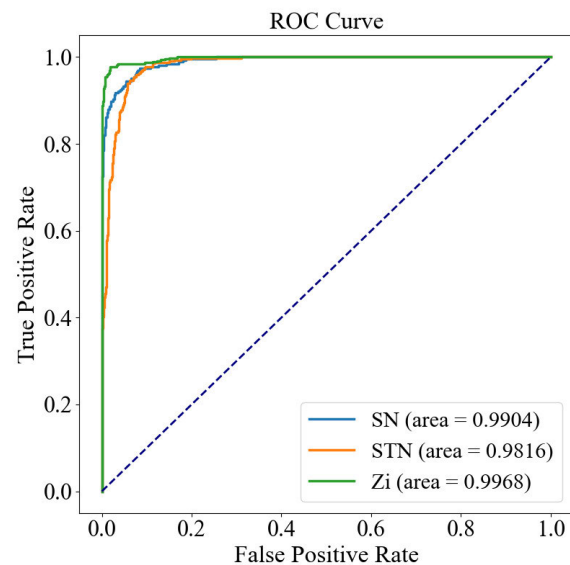


Fig. 11. ROC curves of EEGNet1D trained with expert labels on the test set.

and 24 samples misclassified as Zi. SN classification was also predominantly correct, with 244 samples correctly classified, and its primary error mode was 22 samples misclassified as STN. Overall, the dominant errors arose from confusion between SN and STN, along with partial confusion between STN and Zi. This pattern is consistent with potential overlap in electrophysiological phenotypes among borderline samples from adjacent nuclei.

Fig. 11 shows the ROC curves for the three classes, with AUC values of 0.9968, 0.9904, and 0.9816 for Zi, SN, and STN, respectively. These results indicate that EEGNet1D achieves strong discriminative performance across all three classes, consistent with the relative class separability reflected by the confusion matrix.

Table I summarizes the metric comparison between EEGNet1D and the two supervised baselines on the independent test set. EEGNet1D achieved an overall accuracy of 92.97%,

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT SUPERVISED CLASSIFICATION MODELS ON THE INDEPENDENT TEST SET UNDER EXPERT-CONSENSUS LABEL SETTINGS

Model	Class	Accuracy	Precision	Recall	F1-score
EEGNet1D	Zi		94.99%	98.48%	96.71%
	STN	92.97%	92.53%	87.56%	89.97%
	SN		90.04%	91.73%	90.88%
SVM	Zi		99.33%	96.54%	97.91%
	STN	89.54%	90.76%	79.02%	84.49%
	SN		75.00%	93.61%	83.28%
KNN	Zi		99.10%	94.81%	96.90%
	STN	89.63%	87.06%	83.66%	85.32%
	SN		79.14%	89.85%	84.15%

outperforming SVM and KNN (89.54% and 89.63%, respectively). At the class level, EEGNet1D showed a more pronounced advantage for STN and SN. The $F1$ -score for STN was 89.97%, higher than that of SVM and KNN (84.49% and 85.32%, respectively). The $F1$ -score for SN was 90.88%, also higher than that of SVM and KNN (83.28% and 84.15%). These results suggest that, under supervision with expert-consensus labels, EEGNet1D more effectively distinguishes between nuclei that are prone to confusion than traditional classifiers. This provides a reliable supervised baseline for subsequent pseudo-label training and consistency validation.

C. Training With Pure Clustering Labels and Consistency Evaluation Against the Expert Test Set

To assess the quality of training labels generated directly by clustering algorithms without manual review, this section considered two pure clustering strategies for pseudo-label generation: FCM clustering and spectral clustering. For both methods, feature normalization, PCA-based dimensionality reduction, and cluster fitting were performed exclusively on the training set, after which the clustering outputs were mapped to discrete class labels for EEGNet1D training. The optimal training iteration was selected based on validation-set performance, and final evaluation was conducted on an independent test set using expert-consensus labels as the ground-truth reference.

Fig. 12 shows the confusion matrix of EEGNet1D on the independent test set when trained with FCM-derived pseudo-labels. Overall, the predictions exhibited a clear diagonal pattern, with the Zi class showing the most stable recognition. Specifically, 443 samples were correctly classified, 19 were misclassified as STN, and none were misclassified as SN. The STN class showed bidirectional confusion: 353 samples were correctly identified, whereas 29 were misclassified as SN and 28 as Zi. The SN class was generally well recognized; the primary error mode was 32 samples misclassified as STN, and no samples were misclassified as Zi. Overall, most errors occurred in boundary samples between SN and STN, as well as between STN and Zi.

Fig. 13 shows the corresponding ROC curves. The AUC values for the three classes were 0.9941 (Zi), 0.9796 (SN), and

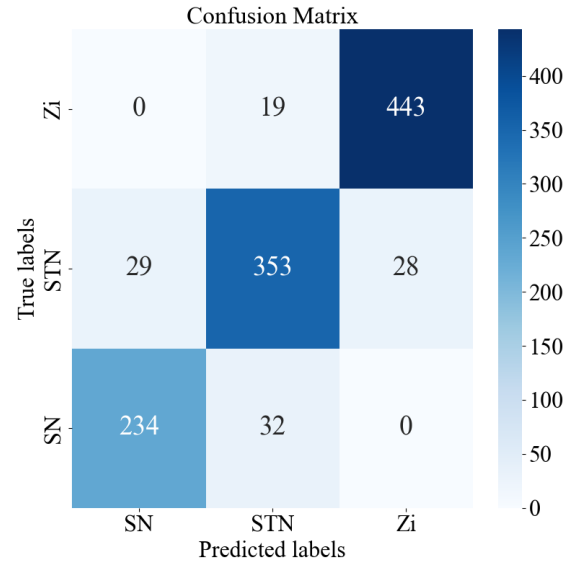


Fig. 12. Confusion matrix of EEGNet1D trained with FCM-based pure clustering labels on the test set.

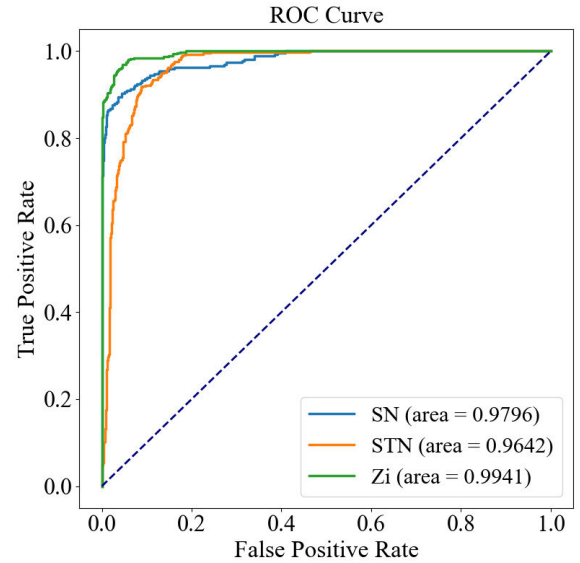


Fig. 13. ROC curves of EEGNet1D trained with FCM-based pure clustering labels on the test set.

0.9642 (STN). These results indicate that, even when trained solely with pure clustering labels, the model retained strong discriminative performance when evaluated against expert-consensus ground truth. However, STN remained relatively more challenging to classify, consistent with the error distribution observed in the confusion matrix.

Table II summarizes the consistency evaluation of the two pure clustering labeling strategies on the independent test set. Overall, FCM-generated pseudo-labels were more robust across all three classes, particularly outperforming spectral clustering for STN and SN. In terms of overall accuracy, FCM achieved 90.51%, whereas spectral clustering achieved 89.10%. At the class level, both methods maintained high consistency for Zi, while the differences became more pronounced for the more challenging STN and SN classes. For STN and SN, FCM achieved $F1$ -scores of 86.73% and

TABLE II

COMPARISON OF CONSISTENCY EVALUATION RESULTS FOR THE PURE CLUSTERING LABEL STRATEGY ON THE INDEPENDENT TEST SET

Clustering algorithms	Class	Accuracy	Precision	Recall	F1-score
FCM	Zi		94.06%	95.89%	94.96%
	STN	90.51%	87.38%	86.10%	86.73%
	SN		88.97%	87.97%	88.47%
Spectral Clustering	Zi		94.04%	95.67%	94.85%
	STN	89.10%	86.86%	82.20%	84.46%
	SN		83.93%	88.35%	86.08%

88.47%, respectively, exceeding those of spectral clustering (84.46% and 86.08%). These results suggest that under pure clustering conditions, FCM produces training labels that are more consistent with expert-defined nuclei than spectral clustering, thereby yielding more stable downstream classification performance.

D. Training With Clustering-Reviewed Labels and Consistency Evaluation Against the Expert Test Set

To achieve a more balanced tradeoff between automated annotation efficiency and label reliability, this section adopts a confidence-based hierarchical review strategy for label generation. First, a clustering model was applied to the training set to generate initial pseudo-labels, with membership degrees or posterior probabilities serving as confidence metrics. High-confidence samples directly retained their cluster labels, whereas low-confidence samples were submitted for expert review and correction. This process yielded a training label set that balances annotation cost and label quality. EEGNet1D was then trained using these reviewed labels. Consistency evaluation was conducted on an independent test set using expert-consensus labels as the ground-truth reference to quantify improvements over pure clustering annotation and the proximity to expert annotations. To ensure comparability across clustering methods under the same review protocol, two clustering baselines—GMM and HDBSCAN—were further introduced for side-by-side comparison with the FCM-based review strategy.

Fig. 14 shows the confusion matrix of EEGNet1D trained with FCM-based label refinement on the independent test set. The Zi class exhibited the most stable recognition, with 453 samples correctly classified; only nine samples were misclassified as STN, and none as SN. The STN class correctly identified 362 samples, and its primary errors arose from boundary confusion with adjacent nuclei; 22 samples were misclassified as SN and 26 as Zi. The SN class correctly classified 240 samples, with 26 misclassified as STN and none as Zi. Overall, the confusion matrix exhibited a clear diagonal pattern, indicating strong class discrimination under this labeling strategy.

Fig. 15 shows the corresponding ROC curves, with AUC values of 0.9955, 0.9885, and 0.9750 for Zi, SN, and STN, respectively. These results indicate that, under the labeling scheme with re-evaluation, the model retained strong

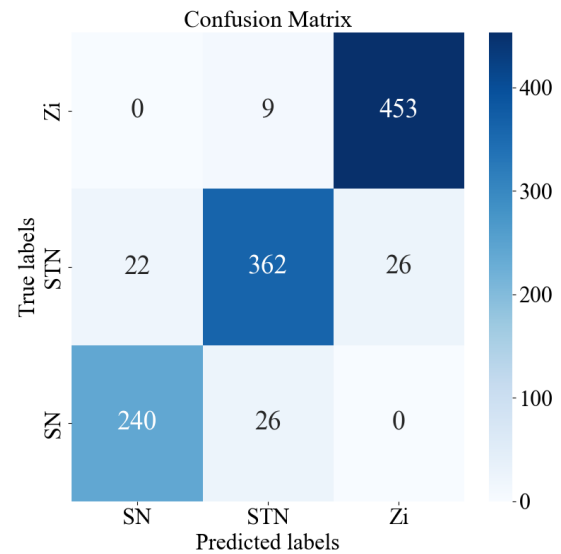


Fig. 14. Confusion matrix of EEGNet1D trained with FCM-based clustering-reviewed labels on the test set.

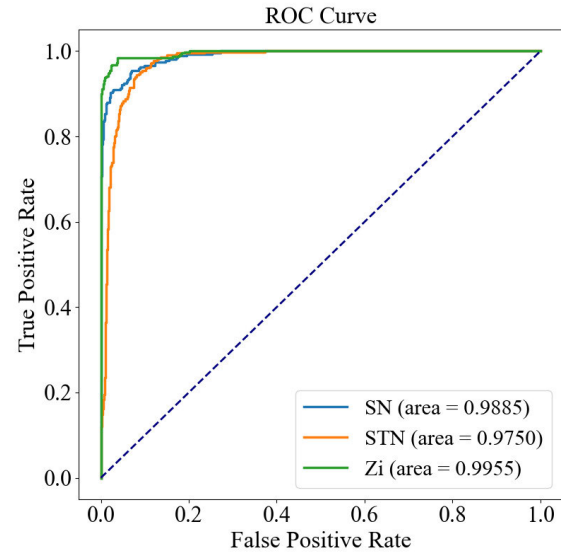


Fig. 15. ROC curves of EEGNet1D trained with FCM-based clustering-reviewed labels on the test set.

discriminative performance across the three classes. Although STN remained relatively more challenging to classify, the training labels generated by the clustering-based re-evaluation strategy still exhibited good reliability and consistency.

Table III summarizes the consistency evaluation of clustering-review labeling strategies based on FCM and GMM on the independent test set. In terms of overall accuracy, the FCM-based reviewed labels achieved 92.71%, exceeding the GMM-based reviewed labels (88.75%). At the class level, FCM showed more balanced performance across all three classes: Zi achieved precision, recall, and $F1$ -score of 94.57%, 98.05%, and 96.28%, respectively; STN achieved 91.18%, 88.29%, and 89.71%; and SN achieved 91.60%, 90.23%, and 90.91%. In contrast, GMM exhibited weaker consistency for STN and SN. For STN, precision, recall, and $F1$ -score were 82.34%, 87.56%, and 84.87%, and for SN, they were 93.15%, 76.69%, and 84.12%. Furthermore, HDBSCAN produced a

TABLE III

COMPARISON OF CONSISTENCY EVALUATION RESULTS FOR THE CLUSTERING-REVIEWED LABEL STRATEGY ON THE INDEPENDENT TEST SET

Clustering algorithms	Class	Accuracy	Precision	Recall	F1-score
FCM	Zi		94.57%	98.05%	96.28%
	STN	92.71%	91.18%	88.29%	89.71%
	SN		91.60%	90.23%	90.91%
GMM	Zi		92.55%	96.75%	94.60%
	STN	88.75%	82.34%	87.56%	84.87%
	SN		93.15%	76.69%	84.12%
HDBSCAN	—	—	—	—	—

TABLE IV

DISTRIBUTION OF SAMPLE CONFIDENCE LEVELS UNDER THE CLUSTERING COMBINED WITH EXPERT REVIEW STRATEGY

Confidence		≥ 0.99	$> 0.3 \text{ and } < 0.99$	≤ 0.3
Sample size	FCM	4327	483	0
	GMM	4708	102	0

substantial number of noisy points and unassigned samples in this dataset, resulting in an excessively high proportion of samples requiring manual review. Consequently, it was difficult to form a complete and trainable three-class label set, and HDBSCAN was therefore excluded from the classification-training comparison in this section.

Compared with the fully expert-labeled setting, the model trained with FCM-based reviewed labels achieved an overall accuracy of 92.71% on the independent test set while substantially reducing manual annotation requirements. As shown in Table IV, under this strategy, 4327 samples met the confidence threshold and were directly assigned automatic labels, whereas only 483 samples entered the expert review process, corresponding to a manual intervention rate of approximately 10%. Compared with the expert-labeling approach that requires annotation of all samples, this strategy maintained high consistency while improving annotation efficiency. Its performance exceeded that of pure clustering-based labeling, indicating that confidence-based stratification and targeted review can alleviate category confusion in borderline samples. Under the same review mechanism, although GMM required review for only 102 samples, resulting in a lower manual intervention rate, its overall accuracy (88.75%) was markedly lower than that of the FCM-based review strategy. This suggests that fewer reviews do not necessarily translate into higher label quality. In contrast, FCM produced more consistent labels while maintaining acceptable review costs.

In summary, the FCM-based clustering re-evaluation strategy achieves a favorable balance between annotation cost and label quality. First, the approach targets only a small subset of low-confidence samples for re-evaluation, thereby substantially reducing manual effort. Second, its overall performance on the independent test set approached that

obtained with expert labels while outperforming pure clustering labels, indicating that the re-evaluation mechanism improves pseudo-label quality. Third, the model maintained stable discriminative performance for Zi and SN. Although some boundary confusion persisted for STN, it was alleviated compared with fully automated clustering. Overall, this strategy provides more reliable training labels with reduced labor cost, suggesting strong potential for clinical use and broader adoption.

V. DISCUSSION

Current studies utilizing MER signals to assist in localizing target nuclei during DBS for PD have primarily focused on binary classification between the STN and non-STN regions. However, such coarse-grained classification strategies are insufficient to meet the clinical demands for precise intraoperative target identification. To address this limitation, the present study extends the binary classification task into a three-class classification framework that distinguishes between the Zi, STN, and SN. This fine-grained approach not only improves the model's generalization ability under complex electrophysiological conditions but also provides more reliable decision support for precise intraoperative navigation during DBS surgery.

To achieve both accurate labeling and efficient classification, this study proposes a novel three-class MER signal classification strategy that integrates supervised feature selection with unsupervised clustering. Through a four-stage process—feature screening, automatic labeling, manual review, and classification validation—the strategy enables comprehensive comparison among unsupervised, hybrid, and expert-labeling approaches. The overall framework operates as follows. First, the random forest algorithm is used to evaluate the importance of the original features and to select a subset of highly discriminative ones. This step enhances the stability and discriminative power of the subsequent clustering process. The selected features are then input into the FCM algorithm for initial clustering, and confidence levels are computed based on sample membership scores. For samples with confidence levels below a predefined threshold, expert review is introduced to correct potential label errors and improve final labeling quality. By retaining the flexibility of unsupervised methods while incorporating supervised feature refinement and selective manual intervention, this strategy effectively addresses the performance degradation often observed in high-noise environments and significantly enhances the reliability of MER signal classification.

In addition, this study systematically compares classification performance across three different labeling strategies: expert labeling, pure clustering labeling, and clustering combined with expert review. The results show that while the expert-labeling group achieved slightly higher accuracy, it is highly labor-intensive, with low annotation efficiency and limited scalability. The pure clustering group demonstrates a high degree of automation but performs poorly in boundary recognition, leading to more frequent misclassifications—particularly in anatomically adjacent regions. The FCM-based clustering-review label group significantly reduces manual intervention

while maintaining overall accuracy close to expert-consensus labels. Only approximately 10% of low-confidence samples require expert review, thereby improving annotation efficiency. This hybrid mechanism of automatic labeling supplemented by targeted manual review offers a practical and scalable solution that balances efficiency and accuracy and effectively reduces the burden of manual annotation in neural signal classification tasks.

In summary, the three-class MER signal classification strategy proposed in this study—based on the integration of supervised feature selection and unsupervised clustering—offers a novel and efficient solution for neural signal classification and label generation. It demonstrates strong robustness and scalability and maintains high labeling efficiency and classification accuracy, making it particularly suitable for real-time intraoperative localization and large-scale neural signal processing. Future research could explore the integration of this framework with multimodal data sources, such as LFPs, neuroimaging, and electrode localization data, to further enhance the precision of multinuclei identification. At the same time, more intelligent confidence-modeling strategies can be introduced to evolve the current fixed-threshold mechanism into an adaptive, engineering-oriented optimization approach. Specifically, thresholds can be differentiated by category and depth position: use a more conservative upper threshold near the STN boundary and a more aggressive threshold in interior regions farther from the boundary. With this mechanism, we expect closed-loop optimization of classification and review, thereby further reducing manual workload while maintaining accuracy. This advancement moves the intelligent neural signal annotation system toward greater automation and clinical utility.

VI. CONCLUSION

The three-class MER signal classification strategy proposed in this study—integrating supervised feature selection with FCM clustering—successfully enables automated labeling and fine-grained distinction of the Zi, STN, and SN. By leveraging the random forest algorithm for feature selection, the clustering performance is effectively enhanced. Furthermore, a manual review mechanism based on sample confidence levels is introduced, which significantly reduces the time cost of manual annotation while ensuring high labeling quality and classification accuracy. This hybrid strategy holds strong application potential in DBS for PD. It provides an efficient and practical technical pathway for rapid labeling and accurate localization of MER signals, and is expected to enhance the intelligence, precision, and clinical utility of intraoperative navigation systems.

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Xiang Lu received the B.S. degree in electronic information engineering and the M.S. degree in control engineering from Shandong University of Science and Technology, Qingdao, China, in 2004 and 2008, respectively, where he is currently pursuing the Ph.D. degree in mine detection and monitoring systems and engineering.

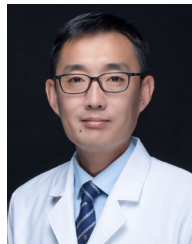
From 2015 to 2016, he was a Visiting Scholar with the College of Instrumentation and Electrical Engineering, Jilin University, Changchun, China.

His research interests include image processing and analysis, medical signal processing, pattern recognition, computer vision, and intelligent perception systems based on system-on-chip (SoC) collaborative design.



Xuexin Du received the B.E. degree in electronic information science and technology from Yantai Institute of Technology, Yantai, Shandong, China, in 2024. He is currently pursuing the M.E. degree in electronic information with Shandong University of Science and Technology, Qingdao, Shandong, China.

His research interests include signal processing, machine learning, and image processing.



Peng Lun received the Doctor of Medicine (M.D.) degree from Qingdao University, Qingdao, China, in 2014.

He is currently an Associate Chief Physician with the Department of Neurosurgery, Affiliated Hospital of Qingdao University, Qingdao, China. Previously, he was a Visiting Scholar at the Rush University Medical Center, Chicago, IL, USA. He has published more than 20 papers in national and international journals. His clinical expertise includes the diagnosis and treatment of movement disorders such as Parkinson's disease, dystonia, and essential tremor, and the diagnosis and treatment of medically refractory epilepsy.

Mr. Lun is a member of the Neuroregulation and Neuroelectrophysiology Committee of Shandong Geriatric Hospital Association and the Second Committee of Parkinson's Disease and Movement Disorders of Shandong Geriatrics Society.



Yande Ren received the Doctor of Philosophy (Ph.D.) degree in medical imaging and nuclear medicine from Guangxi Medical University, Nanning, Guangxi, China, in 2015.

He is currently with the Department of Radiology, Affiliated Hospital of Qingdao University, Qingdao, China. In 2022, he was appointed as a Chief Physician and a Professor. His primary research focus centers on investigating functional and structural brain alterations in neurological and psychiatric disorders using multimodal brain imaging combined

with artificial intelligence (AI). By developing interpretable and generalizable AI models, his work aims to elucidate the multilevel pathogenesis of these disorders and explore objective, accurate methods for diagnostic identification and treatment efficacy evaluation, and ultimately contributing to clinically applicable tools.